

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A colleague fits a linear model to a system that has a tipping point and reports that it “works” on the data they have. Using this chapter, explain what is guaranteed to go wrong, and where.

2. Two systems with completely different mechanisms obey the same differential equation near equilibrium. In terms of linearization, why is this common rather than surprising? What does it say about the flagship example (decay, populations, capacitors) from Chapter 1?

3. You are told a system’s phase space is bounded and its only equilibrium is unstable. What kind of long-term behavior can you *predict* must exist? (Hint: the trajectory cannot leave, and it cannot rest at the equilibrium.)

4. “We have the exact equations, so we can predict the system’s future.” For what class of systems is this inference false even in principle, and why?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
